

Admny ML, DL, NLP
course
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(1)

Sec-38 Decision Tree
Classifier &
Regressor.

lec 1 : Introduction

lec 2 : Entropy &
Gini Impurity

=> Introduction to
Decision Tree Classifier

Decision Tree Classifiers



(A) Entropy and Gini Index
↳ purity split

(b) Information Gain →
features to select for

(ii) Information given
→ features to
select for DT
construction.

~~For~~ For example:-

age = 14

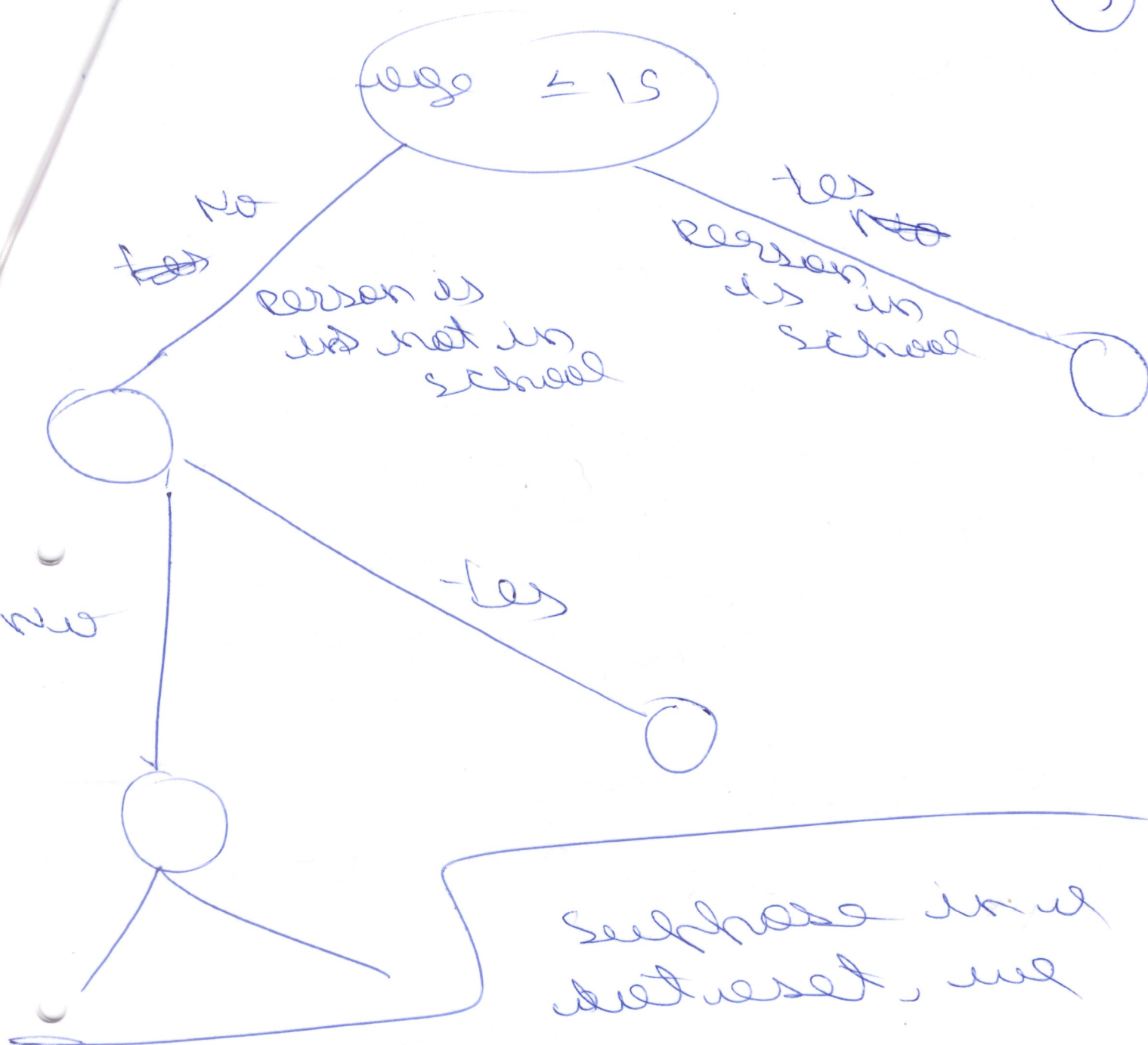
if (age $\leq$ 15):

print("The person
is in school")

elif (age > 15 and age $\leq$ 21):
print("The person
may be in college")

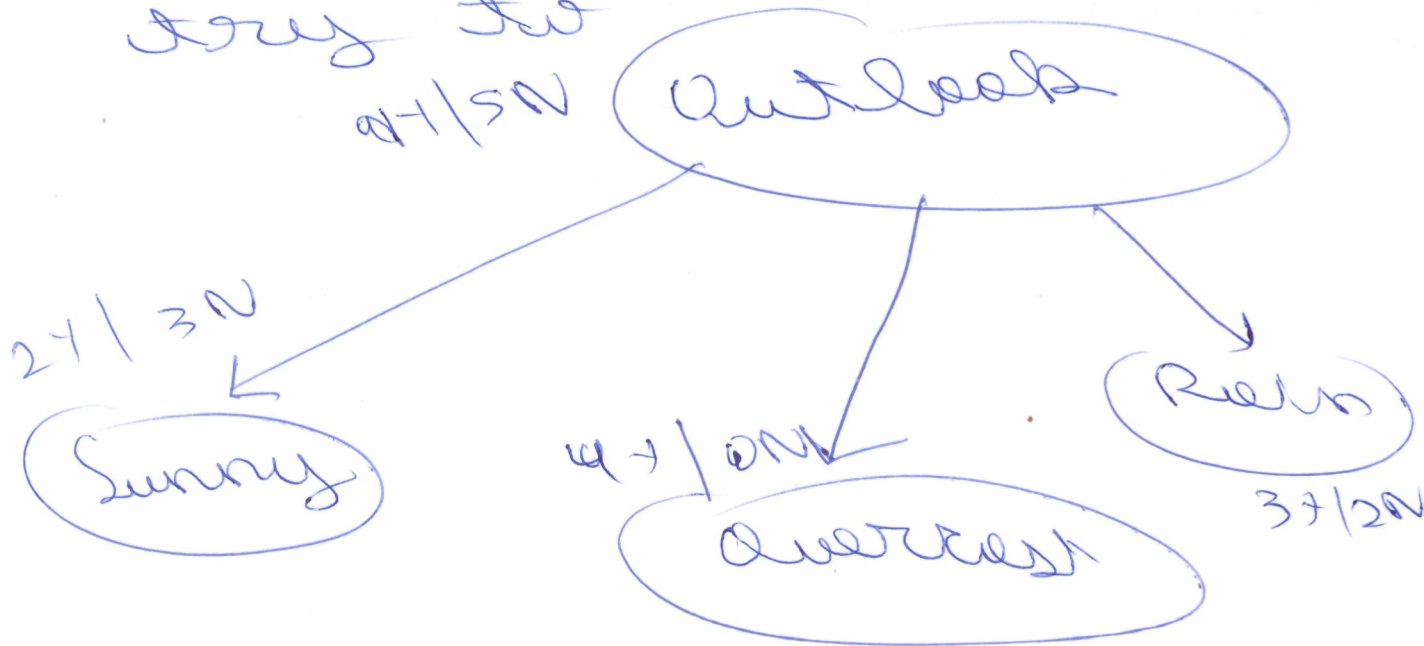
else:

print("The person
has passed the
college").



Suppose it is
dataset, we

try to
at/50



Entropy $\rightarrow$ Pure or Impure
split

$\rightarrow$ Entropy

$\rightarrow$ Gini Impurity

(2) what factor you
need to select best
splitting $\rightarrow$
Information gain

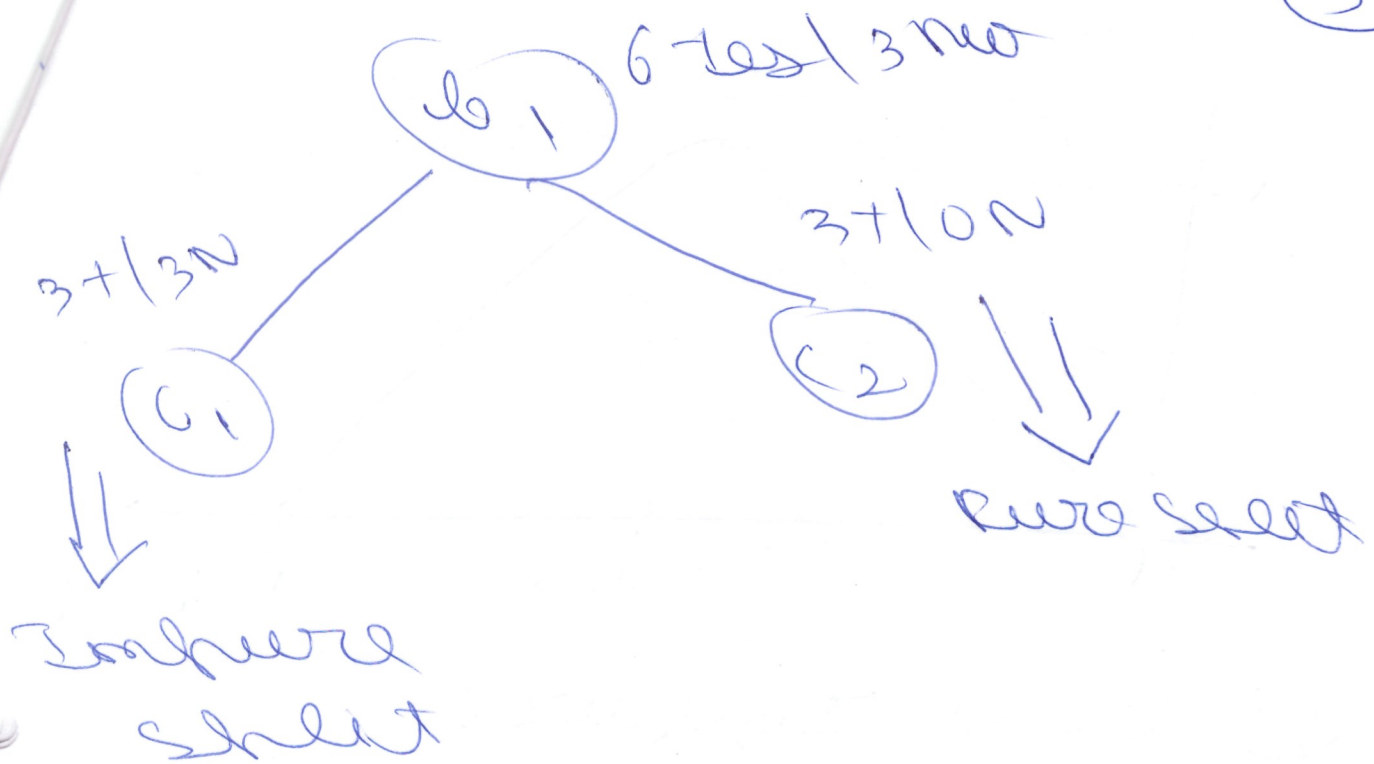
(1) ~~Entropy~~ Entropy

$$H(S) = -p + \log_2 p - p - \log_2 p$$

(2) Gini Impurity:

$$GI = 1 - \sum_{i=1}^n (p_i)^2$$

(5)



$$\begin{aligned}
 H(c_1) &= -p + \log_2 p - p - \log_2 p \\
 &= \frac{-3}{8} \log_2 \frac{3}{8} - \frac{3}{8} \log_2 \frac{3}{8}
 \end{aligned}$$

$\Rightarrow (=)$ Impure split

$$H(c_2) = \frac{-3}{3} \log_2 \frac{3}{3} - 0 \log_2 0$$

$$= -1 \log_2 1 (=) 0 \Rightarrow \text{Pure split}$$



⑫ Gini Ungleichheit

$$G_i = 1 - \sum_{i=1}^n (p_i)^2$$

$$= 1 - ((p_1)^2 + (p_2)^2)$$

$$= 1 - \left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right]$$

$= 0.5 \Rightarrow$ Ungleichheit